

Programming Fundamentals

Assignment: 2

Date: 26-03-2024

Instructor: Shazia Rizwan

Total Marks: 30

Deadline: 01-04-2024

Q:1.) Take a number from a user and extract every digit from the value entered. For example, if the user enters 145 then you have to find the factorial of every single number (1! then 4! then 5!).

- a) Add all these factorial answers. If the resultant number is equal to the entered number, then print "expression is equal" otherwise print "expression is not equal.". (The sum of factorials of all digits in 145 is $1! = 1$, $4! = 24$ and $5! = 120 \rightarrow 145$).
- b) Multiply all these factorial answers. If the resultant number is less than and equal to 1000, display the product and display the last two digits of the product. If the product is greater than 1000 and less than 5000 then display the product and display the last digit. Otherwise just print the product.
- c) If the number entered by user is negative, your program should end by printing "Invalid Value entered" [15 marks]

Q:2.) Consider the working of a snake and ladder game. Your program should accept names for player-1 and for player-2. Dice will be rolled for player-1 for his turn and then for player-2 vice versa. Keep a running total for each player until any of them gets a total score of 100. The player's name who gets a total score of 100 first will be printed along with his score with a message "You Win!...Congratulations". The name of other player is also printed along with his score with a message "Better luck! Next time". Your program is controlled by the user input by asking after the turn of both the players. If any of them types "N" it will terminate your program after printing the existing scores and names respectively. Score should not exceed the value 100.

[15 marks]

Note: Submit your code files in classroom for each question and save it with (Q1,Q2). Any type of cheating material will result in cancellation of your assignment. A viva may be conducted after the submission of your assignment.

TA: Ahmed Fida may also be contacted in case of any issue.